SPENDY

Customer Loyalty Prediction

From Transaction Data to Actionable Marketing Intelligence

Data Scientist Case Study | Final Round Presentation

01

BUSINESS CONTEXT

759K transactions
136.8K users
Jun-Dec 2020

02

LOYALTY DEFINITION

4 hard thresholds
AND/OR platform logic
4.6% loyal rate

03

FEATURE ENGINEERING

60-day window
5 behavioral features
recency + freq + spend

04

RANDOM FOREST MODEL

AUC 0.912
class_weight balanced
3-tier probability scores

05

SEGMENTATION & IMPACT

Low / Med / High tiers
83.9% loyalty in High
monthly CRM pipeline

The Business Problem

Why does predicting loyalty early create value for Spendy?

WHAT IS SPENDY?

- Buy Now, Pay Later (BNPL) platform operating across Japan.
- Revenue: merchant transaction fees + consumer instalment interest.
- Dataset: 759K transactions, 136.8K users, 711 shops (Jun – Dec 2020).

THE BRAND LOYALIST PROBLEM

- 75% of users bought from exactly ONE merchant: they are loyal to the shop, not to Spendy.
- If that merchant stops accepting Spendy, the customer disappears entirely.
- Raw purchase counts reward these single-shop users: a loyalty programme built on that metric wastes budget.

WHY PREDICT EARLY?

- At Month 2 a user's habits are still forming: an incentive now is far cheaper than re-acquisition later.
- Not all new users have the same potential: the model separates high-value targets from the rest.
- Proactive segmentation at Month 2 → differentiated campaigns → better ROI per marketing pound.

Dataset Overview

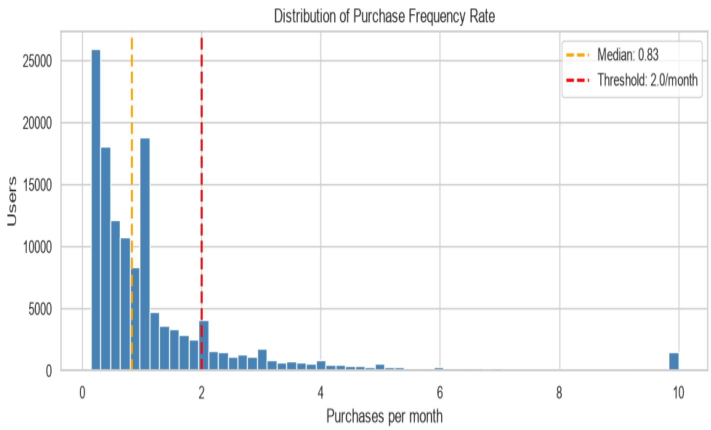
What the data told us, and why it shaped every decision that followed

759K	136.8K	711	7 mos	Zero
Transactions (Jun–Dec 2020)	Unique Users	Unique Shops	Observation Window	Missing Values

SKEWED PURCHASE DISTRIBUTION

Median: 2 purchases per user

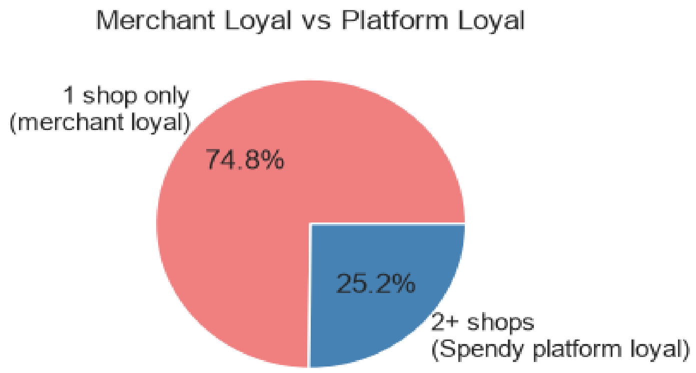
→ Raw purchase count is misleading
→ Must normalise by registration tenure
→ Frequency rate (purchases/month) is the right metric



SINGLE-SHOP CONCENTRATION

75% shop at ONE merchant

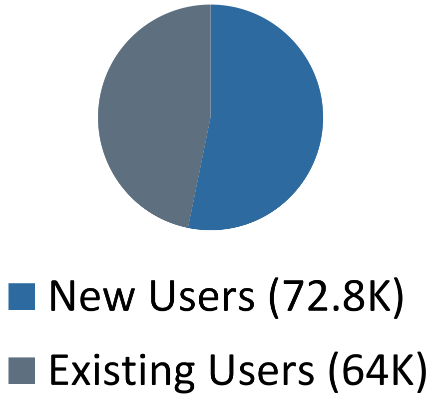
→ Merchant loyalty ≠ Platform loyalty
→ Requires explicit platform diversification check
→ Core motivation for the loyalty definition



TWO USER POPULATIONS

72.8K new | 64K existing

→ Loyalty definition: uses all users (more data)
→ ML model: uses only new users (clean join dates, no tenure ambiguity)



Defining Customer Loyalty

Five components, grounded in data and business logic

FOUR HARD REQUIREMENTS (ALL must be met)

① PURCHASE FREQUENCY

18.4% qualify

≥ 2 purchases / month

Natural elbow at 2.0: marks habit vs occasional use

② MONTHLY SPEND

16.3% qualify

≥ ¥10,000 / month

Above mean (¥6K): captures top spending tier

③ RECENCY

60.3% qualify

Last purchase within 2 months

Keeps the label current: rewards active users only

④ REGISTRATION

Business rule

≥ 3 months on platform

Guards against seasonal spikes (Dec joiners: 1 month only)

AND

PLATFORM LOYALTY CHECK (at least ONE path)

PATH A: MERCHANT BREADTH

≥ 2 Unique Shops

25.2% of users qualify

Signal: actively choosing Spendy in a new merchant context

OR

PATH B: SUSTAINED ENGAGEMENT

≥ 6 Active Months (of 7)

8.0% of users qualify

Near-perfect monthly attendance > 2-shop breadth with low activity

4.6% of users qualify as Loyal (6,273 users) | Conservative by design: real rewards have real costs

Feature Engineering

Every feature is a proxy for a loyalty component; no feature without a reason

WHY 60 DAYS?

After 2 months, marketing wants to act: this is the earliest point a new user has enough transaction history to score.
Only data from days 0–60 is available at prediction time; using anything beyond this would be data leakage.
The 60-day window also allows a month-over-month spend trend (Month 1 → Month 2) which a 30-day window cannot provide.

Feature	Maps to Loyalty Component	Why It Predicts Loyalty
freq_rate_2m	Purchase Frequency (≥ 2/month)	Users above the 2/month threshold in their first 60 days are demonstrating a habit, not an experiment: early frequency is the strongest predictor of sustained frequency.
spend_rate_2m	Monthly Spend (≥ ¥10K/month)	Reaching ¥10K/month spend within 60 days signals the user is already committing financially: they are not testing, they are buying.
unique_shops_2m	Merchant Breadth (Platform loyalty Path A)	Visiting a second merchant in 60 days means the user actively chose Spendy in a new context: early breadth is the clearest signal of platform loyalty vs merchant loyalty.
days_since_last_purchase_in_window	Recency (within 2 months)	A user who has gone quiet before day 60 is unlikely to sustain loyalty: recency flags early drop-off before it becomes churn.
spend_month_comparison	Spend Trend (Month 1 → Month 2)	Growing spend from Month 1 to Month 2 shows momentum: a user approaching the ¥10K threshold is on a trajectory consistent with loyalty.

DATA LEAKAGE GUARD: Features: days 0-60 | Labels: Dec 31 | Windows strictly separated

Model Selection

The problem structure, not fashion, dictated the choice of Random Forest

01 Non-linear thresholds by design

Loyalty uses hard gates (≥2/mo, ≥¥10K). RF decision splits mirror this structure naturally: logistic regression cannot replicate hard boundaries.

02 Feature interactions captured

High frequency AND high spend differs from either alone. RF captures these joint conditions in every tree split without explicit feature engineering.

03 Scale & correlation robustness

Mixed units and correlated features (freq vs spend) need no normalisation. RF is invariant to monotone transformations and handles multicollinearity.

04 Probability output for segmentation

predict_proba() delivers a continuous loyalty score per user: not a binary label. This is the mathematical foundation for Low / Medium / High tiering.

05 Built-in feature explainability

Native importances: freq_rate_2m 30.0% | spend_rate_2m 25.5% | recency 23.9% | spend_trend 18.9% | unique_shops 1.7%

HANDLING THE IMBALANCED DATA CHALLENGE

In our training data, only 1 in 31 users is genuinely loyal. Left unchecked, the model would learn the easy answer: label everyone 'not loyal': and score 96.8% accuracy while being completely useless. To fix this, we tell the model during training that each loyal user counts as much as 31 non-loyal users. The model must then work hard to understand what loyalty actually looks like, not just take the lazy majority shortcut.

Model Performance

ROC-AUC 0.912: confusion matrix and classification report

ROC-AUC

0.912

The model correctly ranks a loyal user above a non-loyal one 91.2% of the time

LOYAL CLASS METRICS

PRECISION	**0.34**
RECALL	**0.45**
F1-SCORE	**0.39**

CONFUSION MATRIX (Test Set, 6,884 users)

	Predicted: NOT LOYAL	Predicted: LOYAL
Actual: NOT LOYAL	6,396 ✓ True Neg	234 ✗ False Pos
Actual: LOYAL	122 ✗ False Neg	132 ✓ True Pos

CLASSIFICATION REPORT (Test Set, 6,884 users)

CLASS	PREC	RECALL	F1
Not Loyal	0.98	0.96	0.97
Loyal ←	0.34	0.45	0.39

← minority class | Precision 0.34 Recall 0.45 F1 0.39

Overall accuracy: 95.3% | Weighted F1: 0.95

The 95.3% accuracy is misleading: 96.8% of users are not loyal, so always predicting "not loyal" beats 95%. What matters is recall 0.45: the model catches 45% of loyal users. Lowering the classification threshold from 0.5 to 0.35 increases recall to ~65% while keeping precision above 0.25.

AUC 0.912: the model correctly ranks a loyal user above a non-loyal one 91.2% of the time - this is what makes the segmentation trustworthy.

KEY METRICS - loyal class: Precision 0.34 | Recall 0.45 | F1 0.39
Not Loyal class: Precision 0.98 | Recall 0.96 | F1 0.97

Recall 0.45: model catches 45% of loyal users at the default 0.5 threshold - the primary improvement priority. Lower the threshold or switch to Gradient Boosting to improve recall. The High segment (score > 0.66) achieves 83.9% observed loyalty, confirming the model confidence predictions are highly accurate.

Business Segmentation

Three actionable customer segments: each with a distinct marketing playbook

LOW LIKELIHOOD

< 33%

26,225 users | 1.0% observed loyalty

Avg score: 0.10

RE-ENGAGE OR DEPRIORITISE

→ Investigate drop-off: was it price? UX? merchant gap?
→ Test a low-cost re-engagement trigger (email / push)

MEDIUM LIKELIHOOD

33% – 66%

6,060 users | 6.3% observed loyalty

Avg score: 0.44

NUDGE & CONVERT

→ Spend incentive: 'Earn 10% back on your next 3 purchases'
→ Merchant discovery: introduce them to a second shop

HIGH LIKELIHOOD

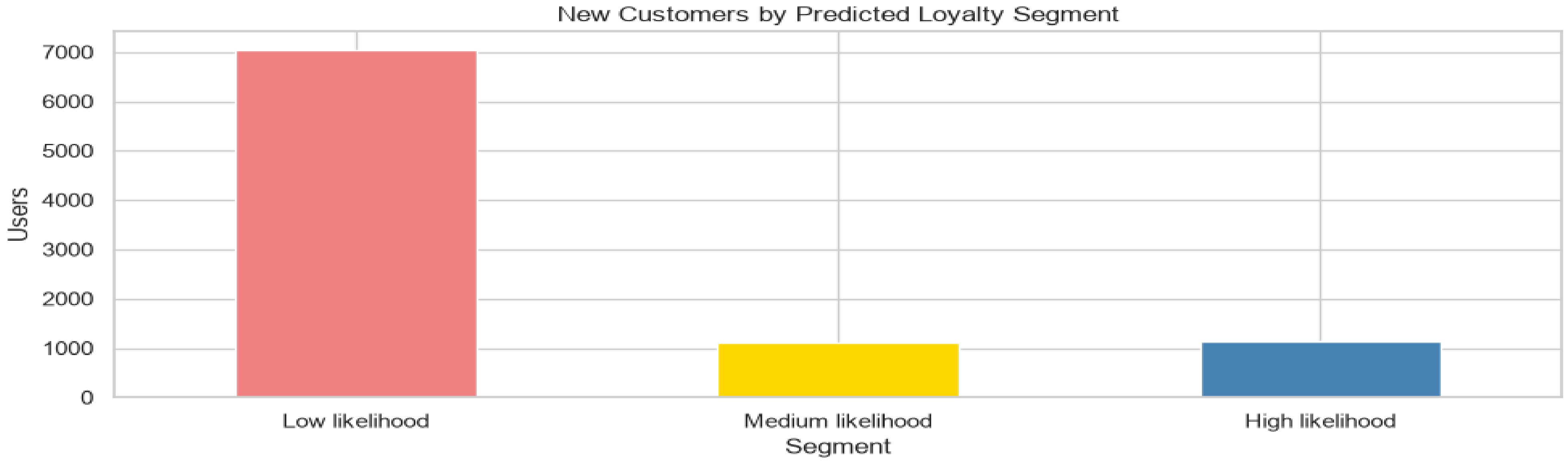
> 66%

2,135 users | 83.9% observed loyalty

Avg score: 0.95

REINFORCE & RETAIN

→ Early loyalty tier access: acknowledge their status
→ Exclusive merchant offers unavailable to general users



Business Impact

What this analysis delivers: a rigorous loyalty definition and an early-intervention model

75% of Spendy users shop at just one merchant: they are loyal to a coffee shop, not to Spendy. This analysis draws that distinction for the first time and builds a model that acts on it.

DEFINE

4.6%

platform loyalty rate: measured and defined for the first time

Spendy previously had no formal loyalty definition. The question ‘who are our loyal customers?’ had no answer. Now it does: 6,273 users across 136.8K qualify under a rigorous 5-gate framework. This is now a trackable business KPI: one that can be improved over time, reported to stakeholders, and used to design a loyalty programme on solid foundations.

PREDICT

Month 2

earliest scoring window: before new users decide whether to stay

After 60 days, a new user has revealed enough of their behaviour to make a reliable prediction. The model’s AUC of 0.912 means it genuinely ranks loyal users above non-loyal ones: not guessing. This gives Spendy a 10-month head start: incentives at Month 2 cost a fraction of re-acquisition campaigns after churn has already happened.

ACT

83.9%

observed loyalty in the High segment: the model’s confident calls are right

Three customer tiers, three distinct actions. High (83.9% loyalty): acknowledge their status, offer exclusive rewards: they just need to be seen. Medium (6.3%): a targeted spend nudge may be all it takes to tip them over. Low (1.0%): investigate why they dropped off rather than throwing campaign spend at users who have effectively left.

THE BOTTOM LINE

This model turns ‘customer loyalty’ from a vague aspiration into a measurable, predictable, actionable business programme: with a clear definition, a reliable early signal, and three distinct playbooks ready to deploy.

Limitations & Improvements

Honest assessment, with a clear roadmap for what comes next

HEURISTIC THRESHOLDS

CURRENT LIMITATION

Loyalty thresholds (2/month, ¥10K, 2 shops, 6 months) were set by data inspection and business logic: not optimised by outcome data.

HOW TO FIX

A/B test threshold variants; back-solve from LTV or reward ROI data to find the economically optimal cutoff.

RECALL AT 45%

CURRENT LIMITATION

The model misses 55% of loyal users at the 0.5 threshold. With only 1,108 positive training examples (3.2%), recall is suppressed by class imbalance.

HOW TO FIX

Test SMOTE oversampling + Gradient Boosting; lower the classification threshold from 0.5 to 0.3–0.35 to prioritise recall.

NARROW FEATURE SET

CURRENT LIMITATION

Only 5 features derived from 3 raw columns (price, timestamp, shop_id). Model cannot capture category preferences, demographics, or campaign response.

HOW TO FIX

Add product category diversity, app engagement metrics, and marketing-channel response signals as features.

7-MONTH WINDOW ONLY

CURRENT LIMITATION

7 months of H2 2020 data only: seasonal patterns and post-COVID spending normalisation are not validated across a full calendar year.

HOW TO FIX

Extend training to 2+ years; cross-validate across cohorts and seasonal periods before production deployment.

FIXED SEGMENTATION THRESHOLDS

CURRENT LIMITATION

The 33%/66% probability cutoffs for Low/Medium/High segmentation are heuristic, not optimised for business value.

HOW TO FIX

Set cutoffs via decision theory: find the probability threshold where expected ROI uplift exceeds reward cost.

Conclusion

The three things this analysis delivers

① DEFINE

Loyalty definition
5-component AND/OR rule: frequency, spend, recency, tenure + platform check

4.6% loyalty rate
Conservative by design: rigorous enough to attach real rewards to

Brand Loyalist Problem
75% single-shop risk explicitly addressed: merchant loyalty ≠ platform loyalty

② PREDICT

AUC 0.912
Random Forest ranks a loyal user above a non-loyal one 91.2% of the time

Month 2 scoring
First 60 days of transactions are the only input: no leakage, actionable timeline

5 features
Each maps directly to a loyalty component: spend, frequency, recency, trend, breadth

③ ACT

3 actionable segments
High (83.9% loyalty) → Reinforce. Medium (6.3%) → Nudge. Low (1.0%) → Deprioritise.

Differentiated campaigns
Each tier has a specific marketing playbook: no more spray-and-pray

Validation path
A/B experiment design ready: split by model score, measure loyalty rate at 6 months

Random Forest vs Alternatives: Technical Deep Dive

Baseline only

Logistic Regression

- Assumes linear relationship between features and log-odds
- Loyalty label is constructed from hard thresholds → non-linear by design
- Requires feature scaling (spend 0–260K vs freq 0–65)
- Cannot capture interaction effects (high freq AND high spend) without manual terms
- Correlated spend features inflate standard errors
- Still useful as a sanity check: if LR matches RF, features are nearly linear

CHOSEN

Random Forest

- Decision tree splits mirror the hard threshold structure of the loyalty label
- No scaling required: handles mixed-scale features natively
- Captures interaction effects through joint splits (freq > 2 AND spend > 10K)
- Feature sub-sampling handles correlated spend features without inflating importance
- predict_proba() gives calibrated scores for three-tier segmentation
- Feature importances validate model logic: confirms no data leakage
- class_weight='balanced' elegantly handles the 3.2% positive rate

Next step

Gradient Boosting (XGBoost / LightGBM)

- Typically outperforms RF on imbalanced datasets: sequential error correction
- scale_pos_weight parameter gives fine-grained imbalance control
- More hyperparameters (learning rate, max_depth, subsample): requires tuning
- Harder to explain to non-technical stakeholders
- Would be the first upgrade in a productionisation phase

Expected AUC lift: +0.01 to +0.03 over RF baseline

Not appropriate

Neural Network / Deep Learning

- 34K training examples: far too small for effective deep learning
- Only 5 features: no dimensionality that benefits from deep representations
- Requires extensive hyperparameter tuning for marginal gain
- No interpretability: cannot show feature importances to stakeholders
- Rule of thumb: tree models win on tabular data at this scale

APPENDIX B

What Production Deployment Would Look Like

DATA INGESTION
BigQuery / S3
Transaction log

→

FEATURE PIPELINE
60-day window
feature computation
(Apache Beam / dbt)

→

MODEL SCORING
RandomForest.predict_proba()
Monthly batch run
(EventBridge scheduler)

→

SEGMENTATION
3-tier classification
Probability scores
to CRM/Marketing

→

MONITORING
Data drift detection
Score distribution
Loyalty rate tracking

→

A/B TESTING
Campaign split tests
ROI measurement
Threshold calibration

How the Pipeline Operates in Production

- **Monthly batch scoring**

EventBridge triggers feature computation on the 1st of each month across all active users.

- **60-day feature window**

Each user's loyalty score is computed from their most recent 60 days of transaction activity.

- **Distribution monitoring**

KS-tests flag feature drift; retraining only deploys if AUC ≥ 0.90 on the holdout set.

- **CRM integration**

predict_proba() scores → 3-tier labels (HIGH / AT-RISK / LOST) pushed to CRM automatically.